Compression Strategies for High-Throughput X-Ray CT of Natural History Collections

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1 Introduction

The *NaturArv3D* pilot project aims to digitize the vertebrate collection at Natural History Museum Denmark (NHMD) in collaboration with the 3D Imaging Centre, Technical University of Denmark (3DIM). As necessitated by the scale of the collection, a custom high-throughput X-ray µCT scanner with a robotic sample handling system is being developed at 3DIM with a designed capacity of **500 samples per week**. Due to the high resolution of the reconstructed volumes, the project is expected to produce approximately **1 PB of data annually**, which presents significant challenges for network infrastructure and data storage that may be mitigated by data compression.

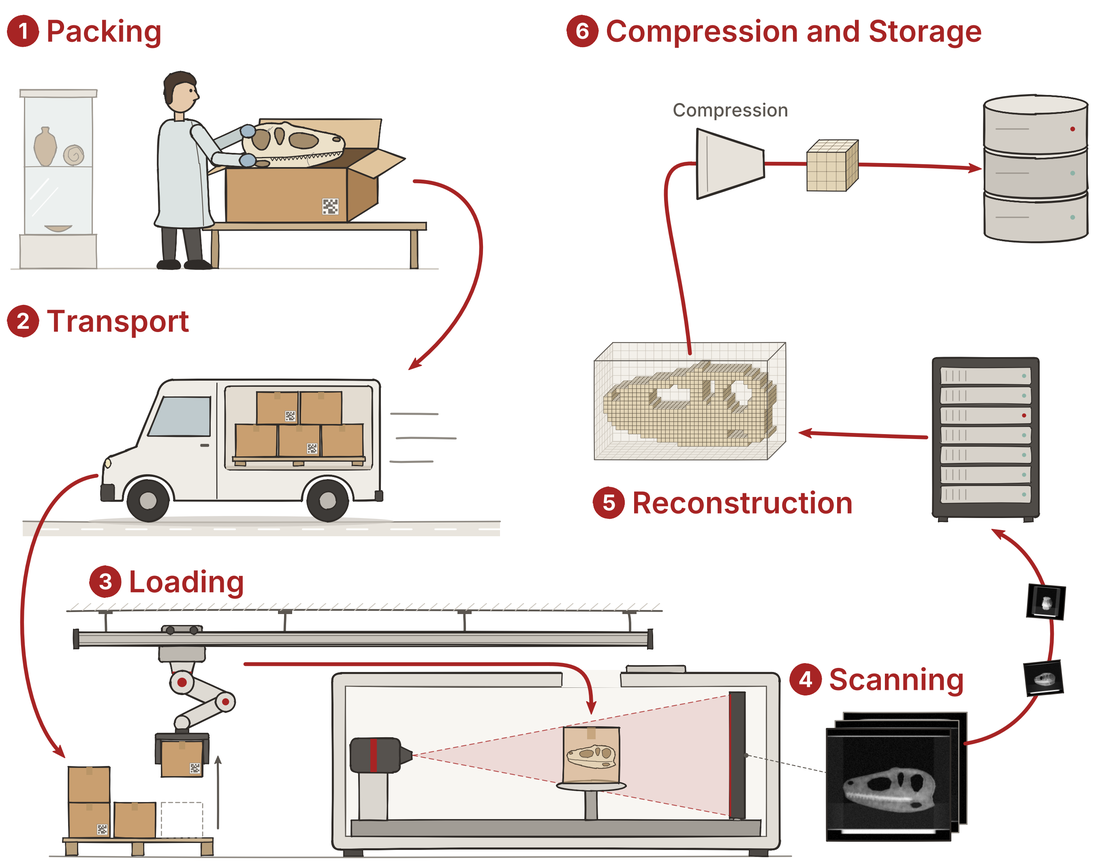


Figure 1: The proposed *NaturArv3D* pipeline for high-throughput X-ray µCT of museum artefacts.

2 Methods

An initial survey of software support across container formats revealed that the DICOM format as the only widely supported format featuring lossy compression (JPEG 2K/LS/XL), which is generally required to acheieve compression ratios above $2:1$ [1]. Uncompressed TIFF stacks provide a baseline for comparison, while the OME-Zarr format with a vast selection of codecs were studied to gauge the performance of modern, less widely supported formats.

A set of TODO representative 16-bit unsigned integer and 32-bit float volumes were compressed for TODO combinations of container format and compression configurations while recording their (de)compression speeds and compression ratio (CR) against the 16-bit uncompressed baseline. Additionally, the fidelity was assessed using full-reference measures (PSNR, SSIM, MS-SSIM).

3 Results

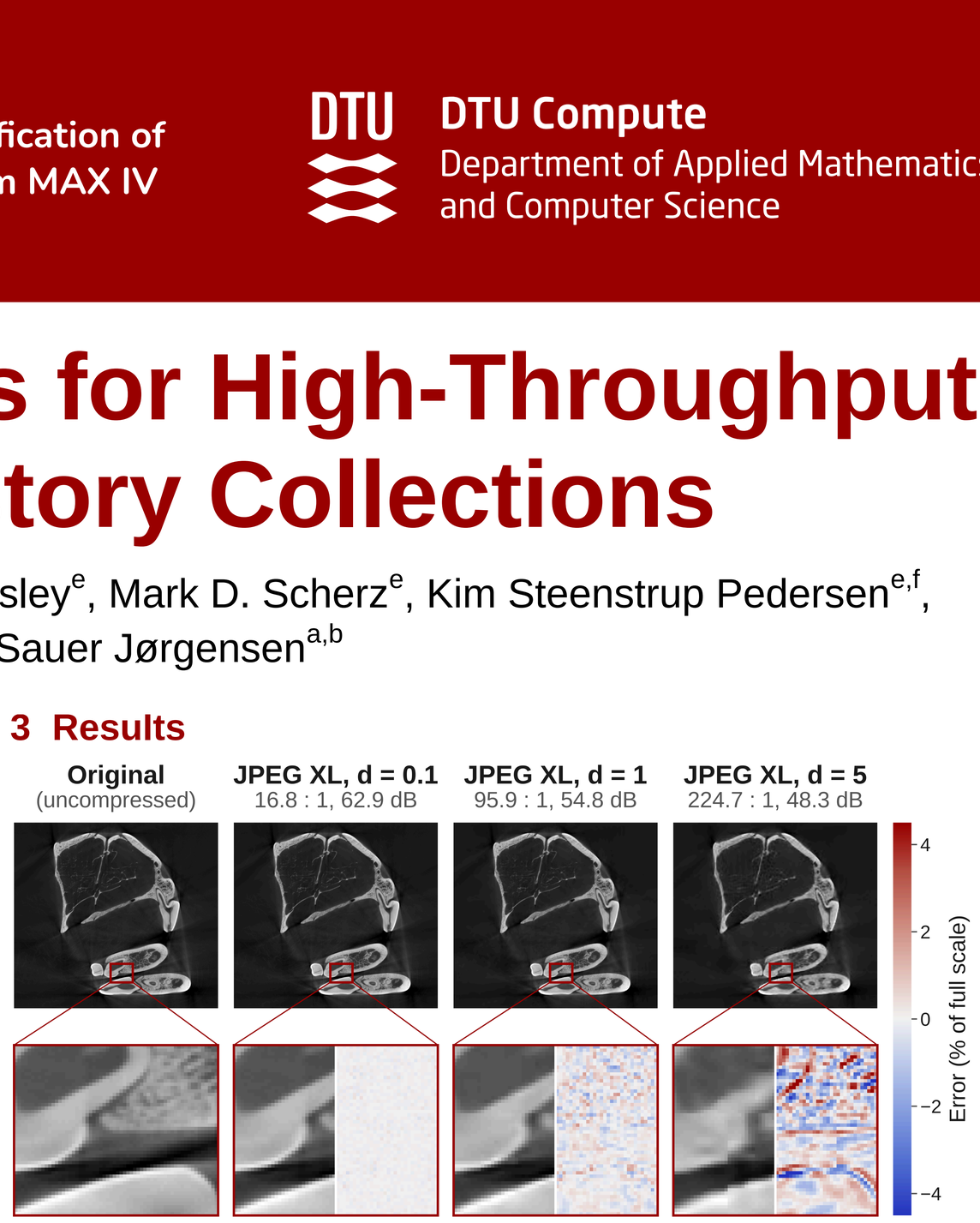


Figure 2: Visualisation of errors in the reconstructed volume for JPEG-XL compression (`effort=7`) *across* the compression planes. Higher distances d indicate greater error bound. Compression with $d = 1$ gives acceptable fidelity with $\mathrm{CR} = 95.6$, corresponding to 98.95% space saving.

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4 Conclusion

5 TODO

- Compression ratio vs fidelity plot
- Error visualization for JPEG-XL
- Cost matrix?

6 References

[1] F. Marone, J. Vogel, and M. Stampanoni, “Impact of Lossy Compression of X-ray Projections onto Reconstructed Tomographic Slices,” *Journal of Synchrotron Radiation*, vol. 27, no. 5, pp. 1326–1338, Sep. 2020, doi: 10.1107/S1600577520007353.

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